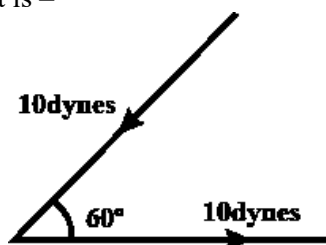


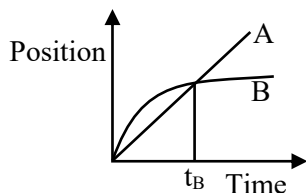
PHYSICS

1. In a book, the answer for a particular question is expressed as $b = \frac{ma}{k} \left[\sqrt{1 + \frac{2k\ell}{ma}} \right]$ here 'm' represents mass, 'a' represents acceleration, 'ℓ' represents length. The unit of b should be
 (1) m/s (2) m/s²
 (3) meter (4) second
2. $\alpha = \frac{F}{V^2} \sin(\beta t)$ (here V = velocity, F = force, t = time) Find the dimension of α and β -
 (1) $\alpha = [M^1 L^1 T^0]$, $\beta = [T^{-1}]$
 (2) $\alpha = [M^1 L^1 T^{-1}]$, $\beta = [T^1]$
 (3) $\alpha = [M^1 L^1 T^{-1}]$, $\beta = [T^{-1}]$
 (4) $\alpha = [M^1 L^{-1} T^0]$, $\beta = [T^{-1}]$
3. $\int \left(\sin x + 2e^x + \frac{4}{x} \right) dx = ?$
 (1) $\cos x + 2e^x - \frac{4}{x^2} + c$
 (2) $-\cos x + 2e^x - \frac{4}{x^2} + c$
 (3) $-\cos x + 2e^x + 4 \log_e x + c$
 (4) $-\cos x + 2e^x - 4 \log_e x + c$
4. The velocity of water waves may depends on their wavelength λ , the density of water ρ and the acceleration due to gravity g. The method of dimensions gives the relation between these quantities where k is a dimensionless constant
 (1) $v^2 = k\lambda^{-1} g^{-1} \rho^{-1}$
 (2) $v^2 = k g \lambda$
 (3) $v^2 = k g \lambda \rho$
 (4) $v^2 = k \lambda^3 g^{-1} \rho^{-1}$
5. In a certain system of units, 1 unit of time is 5 sec, 1 unit of mass is 20 kg and unit of length is 10 m. In this system, one unit of power will correspond to
 (1) 16 Watt (2) 1/16 Watt
 (3) 25 Watt (4) None of these
6. An experiment measures quantities x, y, z and then 't' is calculated from the data as, $t = \frac{xy^2}{Z^3}$. If percentage errors in x, y and z are respectively 1%, 3%, 2%, then percentage error in 't' is:
 (1) 10% (2) 4%
 (3) 7% (4) 13%
7. A student measures the time period of 100 oscillations of a simple pendulum four times. The data set is 90 s, 91 s, 95 s and 92 s. If the minimum division in the measuring clock is 1 s, then the reported mean time should be:
 (1) 92 ± 1.5 s (2) 92 ± 5 s
 (3) 92 ± 1.8 s (4) 92 ± 3 s
8. Two stones are projected with the same speed but making different angles with the horizontal. Their horizontal ranges are equal. The angle of projection of one is $\pi/3$ and the maximum height reached by it is 102 m. Then the maximum height reached by the other in metre is -
 (1) 336 m (2) 224 m
 (3) 56 m (4) 34 m
9. Suppose that the radius r and area $A = \pi r^2$ of a circle are differentiable functions of t. Equation that relates dA/dt to dr/dt is.
 (1) $\frac{dA}{dt} = \pi r \frac{dr}{dt}$ (2) $\frac{dA}{dt} = \pi r^2 \frac{dr}{dt}$
 (3) $\frac{dA}{dt} = 2\pi r^2 \frac{dr}{dt}$ (4) $\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$
10. Vector $\vec{A}$ is of length 2 cm and is 60° above the x-axis in the first quadrant. Vector $\vec{B}$ is of length 2 cm and 60° below the x-axis in the fourth quadrant. The sum $\vec{A} + \vec{B}$ is a vector of magnitude-
 (1) 2 along + y-axis (2) 2 along + x-axis
 (3) 1 along - x axis (4) 2 along - x axis
11. Three forces P, Q&R are acting at a point in the plane. The angle between P & Q and Q&R are 150° & 120° respectively, then for equilibrium, forces P, Q&R are in the ratio
 (1) 1:2:3 (2) 1:2: $\sqrt{3}$
 (3) 3:2:1 (4) $\sqrt{3}$:2:1

12. Two forces each numerically equal to 10 dynes are acting as shown in the following figure, then their resultant is –



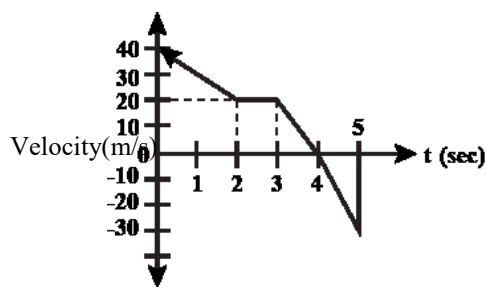
- (1) 10 dynes (2) 20 dynes
(3) $10\sqrt{3}$ dynes (4) 5 dynes
13. If $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors such that $\hat{i} + \hat{j} + \hat{k} = \hat{0}$ then the angle between $\hat{i}$ and $\hat{j}$ is
(1) 30° (2) 60°
(3) 90° (4) 120°
14. The graph shows position as a function of time for two trains running on parallel tracks. Which one of the following statement is true?



- (1) At time t_B , both trains have the same velocity.
(2) Both trains have the same velocity at some time after t_B
(3) Both trains have the same velocity at some time before t_B .
(4) Somewhere on the graph, both trains have the same acceleration.
15. A body sliding on a smooth inclined plane requires 4sec to reach the bottom after starting from rest at the top. How much times does it take to cover one fourth the distance starting from on the top
(1) 1sec (2) 2sec
(3) 0.4sec (4) 1.6sec
16. The displacement time graphs of two particles A and 'B' are straight lines making angles of respectively 30° and 60° with the time axis. If the velocity of A is v_A and that of B is v_B then the value of $\frac{v_A}{v_B}$ is

- (1) $1/2$ (2) $1/\sqrt{3}$ (3) $\sqrt{3}$ (4) $1/3$

17. In the following velocity-time graph of a body, the distance and displacement travelled by the body in 5 second in meters will be-



- (1) 75, 115 (2) 105, 75
(3) 45, 75 (4) 95, 55
18. Mark the correct statement for a particle going on a straight line
(1) if the velocity is zero for a time interval, the acceleration is zero at any instant within the time interval
(2) if the velocity and acceleration have opposite sign, the object is slowing down
(3) if the position and velocity have opposite sign, the particle is moving towards the origin.
(4) All of the above
19. The displacement x of a particle depend on time t as $x = \alpha t^2 - \beta t^3$
(1) particle will return to its starting point after time α/β .
(2) the particle will come to rest after time $\frac{2\alpha}{3\beta}$
(3) the initial velocity of the particle was zero but its initial acceleration was not zero.
(4) All of the above
20. If a ball is thrown vertically upwards with speed 'u', the distance covered during the last 't' seconds of its ascent is:
(1) ut (2) $ut - \frac{1}{2}gt^2$
(3) $\frac{1}{2}gt^2$ (4) $(u + gt)t$
21. A particle moves from the point $(2.0\hat{i} + 4.0\hat{j})$ m, at $t = 0$ with an initial velocity $(5.0\hat{i} + 4.0\hat{j})$ ms^{-1} . It is acted upon by a constant force which produces a constant acceleration $(4.0\hat{i} + 4.0\hat{j})$ ms^{-2} . What is the distance of the particle from the origin at time 2s?

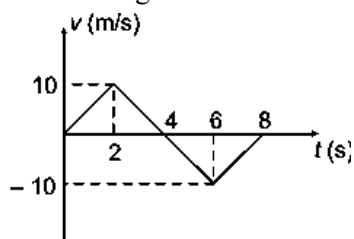
- (1) 15m (2) $20\sqrt{2}$
(3) 5m (4) $10\sqrt{2}$ m

22. **Statement I:** In projectile motion, the angle between the instantaneous velocity and acceleration at the highest point is 180° .

Statement II: At the highest point, velocity of projectile will be in horizontal direction only.

- (1) If both Statement I and Statement II are true and the Statement II is correct explanation of the Statement I.
- (2) If both Statement I and Statement II are true but Statement II is not the correct explanation of Statement I.
- (3) If Statement I is true, but the Statement II is false.
- (4) If Statement I is false but the Statement II is true.

23. Corresponding to velocity-time graph in one dimensional motion of a particle as shown in figure, match the following two columns.



Column - I		Column - II	
(A)	Average velocity between zero second and 4 s	(p)	10 SI units
(B)	Average acceleration between 1 s and 4 s	(q)	2.5 SI units
(C)	Average speed between zero second and 6 s	(r)	5 SI units
(D)	Rate of change of speed at 4 s	(s)	None of the above

- (1) (A→p), (B→q), (C→r), (D→r)
- (2) (A→s), (B→q), (C→s), (D→r)
- (3) (A→r), (B→s), (C→p), (D→p)
- (4) (A→r), (B→s), (C→r), (D→r)

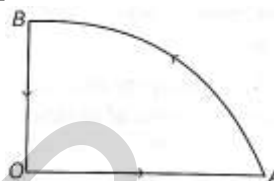
24. Maximum & minimum value of function $y = 3 \sin x + 4 \cos x$ will be respectively:

- (1) 4, -4
- (2) 5, -5
- (3) 7, -1
- (4) 3, -3

25. The displacement x of a particle varies with time t , $x = ae^{-pt} + be^{qt}$, where a , b , p and q are positive constants. The velocity of the particle will

- (1) go on increasing with time
- (2) be independent of p and q
- (3) drop to zero when $p = q$
- (4) go on decreasing with time

26. An object is moving along the path OABO with constant speed, then



- (1) The acceleration of the object while moving along to path OABO is zero
- (2) The acceleration of the object along the path OA and BO is zero
- (3) There must be some acceleration along the path AB
- (4) Both (2) and (3)

27. From a tower of height H , a particle is thrown vertically upwards with a speed u . The time taken by the particle to hit the ground is n times the time taken by it to reach the highest point of its path. The relation between H , u and n is:

- (1) $2gH = n^2u^2$
- (2) $gH = (n-2)^2u^2$
- (3) $2gH = nu^2(n-2)$
- (4) $gH = (n-2)^2u^2$

28. Rain is falling vertically with a speed of 35ms^{-1} . A woman rides a bicycle with a speed of 12ms^{-1} in east to west direction. The direction in which she should hold her umbrella is

- (1) at $\cos^{-1}(0.343)$ with vertical towards east
- (2) at $\tan^{-1}(0.343)$ with vertical towards west
- (3) at $\cos^{-1}(0.343)$ with vertical towards west
- (4) at $\tan^{-1}(0.343)$ with vertical towards east

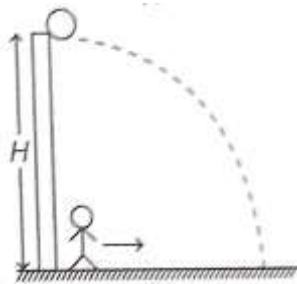
29. A girl riding a bicycle with a speed of 5ms^{-1} towards east direction sees raindrops falling vertically downwards. On increasing the speed to 15ms^{-1} , rain appears to fall making an angle of 45° of the vertical. Find the magnitude of velocity of rain.

- (1) 5ms^{-1}
- (2) $5\sqrt{5}\text{ms}^{-1}$
- (3) 25ms^{-1}
- (4) 10ms^{-1}

30. A particle is moving along the X-axis with its coordinate given by $x(t) = 10 + 8t - 3t^2$. Another particle is moving along the Y-axis with its coordinate as a function of time given by $y(t) = 5 - 8t^3$. At $t = 1$ sec., the speed of the second particle as measured in the frame of the first particle is given as $\sqrt{v}$. Then, v (in m/s) is

- (1) 430 ms^{-1} (2) 580 ms^{-1}
(3) 450 ms^{-1} (4) 380 ms^{-1}

31. A ball is thrown horizontally from the top of a tower of height H with velocity 20 m/s . A man is standing at the foot of tower starts running at the same instant when ball is thrown. What should be the velocity of man to catch the ball before striking to the ground. (Consider height of man negligible.)



- (1) 10 m/s (2) $\sqrt{2gH}$
(3) 20 m/s (4) 30 m/s

32. A fighter plane flying horizontally at an altitude of 1.0 km with speed 720 km h^{-1} passes directly overhead an anti-aircraft gun. At what angle from the vertical should the gun be fired for the shell will muzzle speed 400 ms^{-1} to hit the plane. (Take $g = 10 \text{ ms}^{-2}$)
- (1) 60° (2) 45° (3) 38° (4) 30°

33. Two projectiles having different masses m_1 and m_2 are projected at an angle α and $(90^\circ - \alpha)$ with the same speed from same point. The ratio of their maximum heights is

- (1) $\cot \alpha : \sin \alpha$ (2) $1 : 1$
(3) $\tan^2 \alpha : 1$ (4) $1 : \tan \alpha$

34. **Statement I:** A projectile always follows a parabolic path.

Statement II: In projectile motion, momentum is never zero throughout the motion.

- (1) Statement -1 is True, Statement-2 is True, Statement-2 is a correct explanation for Statement-1.
(2) Statement-1 is True, Statement-2 is True, Statement-2 is Not a correct explanation for Statement-1.
(3) Statement-1 is True, Statement-2 is False.
(4) Statement-1 is False, Statement-2 is True.

35. **Statement I:** At highest point of projectile, possesses velocity only along the horizontal direction.

Statement II: At highest point of projectile, the velocity and acceleration are perpendicular to each other.

- (1) Statement -1 is True, Statement-2 is True, Statement-2 is a correct explanation for Statement-1.
(2) Statement-1 is True, Statement-2 is True, Statement-2 is Not a correct explanation for Statement-1.
(3) Statement-1 is True, Statement-2 is False.
(4) Statement-1 is False, Statement-2 is True.

36. **Statement I:** When an object follows a circular path at a constant speed, the motion of the object is called uniform circular motion.

Statement II: The acceleration of an object in uniform circular motion is always directed away the centre of the circle

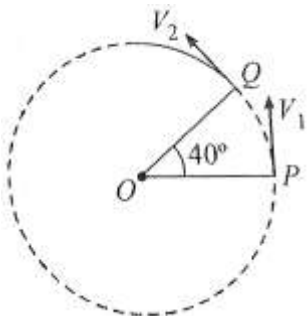
- (1) Statement -1 is True, Statement-2 is True, Statement-2 is a correct explanation for Statement-1.
(2) Statement-1 is True, Statement-2 is True, Statement-2 is Not a correct explanation for Statement-1.
(3) Statement-1 is True, Statement-2 is False.
(4) Statement-1 is False, Statement-2 is True.

37. A football player kicks a ball at an angle of 37° to the horizontal with an initial speed of 15 ms^{-1} . Assuming that the ball travels in a vertical plane, match Column I with Column II. (Given, $\sin 37^\circ = 0.601$; $\sin 74^\circ = 0.961$)

Column-I		Column-II	
(A)	Time at which ball reaches the highest point (in s)	(1)	4.16
(B)	Maximum height reached (in m)	(2)	1.84
(C)	Horizontal range of the projectile (in m)	(3)	0.92
(D)	Time for which ball is in air (in s)	(4)	22.02

Choose the correct answer from the options given below.

- (1) (A)→(3), (B)→(1), (C)→(4), (D)→(2)
 (2) (A)→(2), (B)→(4), (C)→(3), (D)→(1)
 (3) (A)→(1), (B)→(2), (C)→(4), (D)→(3)
 (4) (A)→(4), (B)→(3), (C)→(1), (D)→(2)
38. Water drops fall at regular intervals from a roof. At an instant when a drop is about to leave the roof, the separations between 3 successive drops below the roof are in the ratio:
- (1) 1 : 2 : 3 (2) 1 : 4 : 9
 (3) 1 : 3 : 5 (4) 1 : 5 : 13
39. A particle is moving in a circle of radius r centred at O with constant speed v . The change in velocity in moving from P to Q ($\angle POQ = 40^\circ$) is:

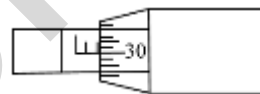


- (1) $2v \cos 40^\circ$ (2) $2v \sin 40^\circ$
 (3) $2v \cos 20^\circ$ (4) $2v \sin 20^\circ$

40. A bird flies for 4 sec with a velocity of $|t - 2| \text{ m/s}$ in a straight line, where $t = \text{time in seconds}$. It covers a distance of :
 (1) 2 m (2) 4 m (3) 6 m (4) 8 m

41. The period of oscillation of a simple pendulum is $T = 2\pi \sqrt{\frac{L}{g}}$. Measured value of L is 1.0 m from metre scale having a minimum division of 1 mm and time of one complete oscillation is 1.95 s measured from stopwatch of 0.01 s resolution. The percentage error in the determination of g will be
 (1) 1.13% (2) 1.03%
 (3) 1.33% (4) 1.30%

42. Least count of main scale of screw gauge is 1 mm and there are 50 equal calibration on circular scale, circular scale move 1 part main scale on 1 rotation of circular scale then zero error of given this diagram will be:



- (1) 2.30 mm (2) -2.30 mm
 (3) 2.60 mm (4) -2.60 mm

43. Unit of electric flux is: (where $\phi = \vec{E} \cdot \vec{A}$ and $\vec{E} = \frac{Q}{\epsilon_0 r^2} \hat{r}$ and also $E = -\frac{dV}{dr}$, where dV is potential difference)
 (1) V-m (2) N-m/C
 (3) V/m (4) C/N-m

44. If vector $\vec{A}$ is added to $\vec{B}$ then the resultant is $5\hat{i} + 3\hat{j}$ and if $\vec{B}$ is subtracted from $\vec{A}$ the resultant is $3\hat{i} - \hat{j}$, then projection of $\vec{A}$ on $\vec{B}$ will be :
 (1) 6 (2) $6/\sqrt{5}$
 (3) $\sqrt{5}/6$ (4) None

45. Vectors A and B include an angle α between them. If $(A + B)$ and $(A - B)$, respectively subtend angles θ_1 and θ_2 with A , then $(\tan \theta_1 - \tan \theta_2)$ is:
 (1) $\frac{B^2 \cos 2\alpha}{A^2 - B^2 \cos^2 \alpha}$ (2) $\frac{-B^2 \cos 2\alpha}{A^2 - B^2 \cos^2 \alpha}$
 (3) $\frac{-B^2 \sin 2\alpha}{A^2 - B^2 \cos^2 \alpha}$ (4) $\frac{B^2 \sin 2\alpha}{A^2 - B^2 \cos^2 \alpha}$

CHEMISTRY

- 46.** How many moles of electron weight 10 g.
- (1) 6×10^{23} (2) $\frac{10^{31}}{9.1}$
 (3) $\frac{10^6}{9.1 \times 6}$ (4) $\frac{10^8}{6 \times 9.1}$
- 47.** Two element X (At. wt 75) & Y (At. wt 16) when combined form a compound in which 75.8 % X is present find empirical formula of compound.
- (1) XY (2) X₂Y
 (3) X₃Y₂ (4) X₂Y₃
- 48.** Mass of H₂O in 1000 kg CuSO₄·5H₂O is (At. wt of Cu = 63.5g)
- (1) 360.5 kg (2) 36.05 kg
 (3) 3605 kg (4) 3.605 kg
- 49.** The energy of second Bohr orbit of the hydrogen atom is -328KJ mol⁻¹, hence the energy of fourth Bohr orbit would be
- (1) -41 (2) -1312
 (3) -164 (4) -82
- 50.** According to the following reaction
 $A + BO_3 \longrightarrow A_3O_4 + B_2O_3$
 The no. of mole of A₃O₄ produced if 2 mole of A and 1 mole BO₃ react together.
- (1) 3 (2) 1 (3) $\frac{3}{8}$ (4) $\frac{2}{3}$
- 51.** Volume of CO₂ obtained by the complete decomposition of 9.85g of BaCO₃ is
- (1) 2.24 L (2) 1.118 L
 (3) 0.84 L (4) 0.56 L
- 52.** M gm of a substance when vaporised occupy a volume of 5.6 litre at NTP. The molecular mass of the substance will be:
- (1) M (2) 2M
 (3) 3M (4) 4M
- 53.** If 30 ml of H₂ and 20 ml of O₂ react to form water, what is left at the end of the reaction
- (1) 10 ml of H₂ (2) 10 ml of O₂
 (3) 5 ml of O₂ (4) 5 ml of H₂
- 54.** Which one of the following set of ions represent the collection of isoelectronic species.
- (1) K⁺, Ca⁺², Sc⁺³, Cl⁻
 (2) Na⁺, Cs⁺², Sc⁺³, F⁻
 (3) K⁺, Cl⁻, Mg⁺², Sc⁺³
 (4) Na⁺, Mg⁺², Al⁺³, Cl⁻
- 55.** Which of the following set of quantum number is correct for an electron in 4f orbital?
- | n | ℓ | m | S |
|-------|---|----|------|
| (1) 4 | 3 | 4 | +1/2 |
| (2) 4 | 4 | -4 | -1/2 |
| (3) 4 | 3 | 1 | +1/2 |
| (4) 4 | 2 | -2 | +1/2 |
- 56.** Which of the following represents 180g of water correctly ?
- (1) 5 mole of H₂O
 (2) 6.023×10^{23} molecules of water
 (3) 1 mole of H₂O
 (4) 6.023×10^{24} molecules of water
- 57.** Which of the following expression represent the spectrum of Balmer series (if n is the principal quantum no. of higher energy level)
- (1) $\bar{\nu} = \frac{R(n-1)(n+1)}{n^2} \text{ cm}^{-1}$
 (2) $\bar{\nu} = \frac{R(n-2)(n+2)}{4n^2} \text{ cm}^{-1}$
 (3) $\bar{\nu} = \frac{R(n-2)(n+2)}{n^2} \text{ cm}^{-1}$
 (4) $\bar{\nu} = \frac{R(n-1)(n+1)}{4n^2} \text{ cm}^{-1}$
- 58.** An electron has principal quantum number 3 the number of its (i) subshell and (ii) orbitals would be respectively –
- (1) 3 & 5 (2) 3 & 7
 (3) 3 & 9 (4) 2 & 5
- 59.** What is the maximum number of electrons, in an atom that can have the following quantum numbers n = 4, m = +1?
- (1) 4 (2) 15
 (3) 3 (4) 6

60. Match List I with List II.

List I		List II	
(a)	1 mol water	(i)	22.4 L
(b)	1 g molecules of O ₂ at STP	(ii)	0.06 mol neutrons
(c)	12.022×10^{23} molecules of N ₂ H ₄	(iii)	18 mL
(d)	0.16 g CH ₄ contains	(iv)	8 g atoms of H

(1) (a)-iii, (b)-i, (c)-iv, (d)-ii

(2) (a)-iv, (b)-i, (c)-iii, (d)-ii

(3) (a)-iii, (b)-iv, (c)-ii, (d)-i.

(4) (a)-i, (b)-iii, (c)-ii, (d)-iv

61. Which of the following will contain maximum number of molecules?

(1) 100L CO₂(g) at STP

(2) 10g of H₂(g) at STP

(3) 1L of H₂O (ℓ)

(4) 1000 g of C₁₂H₂₂O₁₁(s) at STP

62. If 9 moles of O₂ and 14 moles of N₂ are placed in a container and allowed to react according to the equation $3\text{O}_2 + 2\text{N}_2 \rightarrow 2\text{N}_2\text{O}_3$. The reaction proceeds until 3 moles of O₂ remain, how many moles of N₂O₃ are present at that instant?

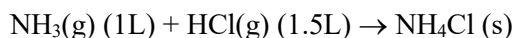
(1) 6

(2) 3

(3) 4

(4) 12

63. What will be volume of the mixture after the reaction



(1) 1.5 L

(2) 0.5 L

(3) 1 L

(4) 0 L

64. Weight of oxygen in Fe₂O₃ and FeO is in the simple ratio for the same amount of iron is :

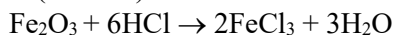
(1) 3 : 2

(2) 1 : 2

(3) 2 : 1

(4) 3 : 1

65. When 20 g Fe₂O₃ is reacted with 50 g of HCl, FeCl₃ and H₂O are formed. The amount of unreacted HCl is (Fe = 56)



(1) 7.375 g

(2) 22.625 g

(3) 80 g

(4) 4.75 g

66. A sample of ammonium phosphate (NH₄)₃PO₄ contains 3 mol of H atoms. The number of mol of O atoms in the sample is:

(1) 0.265

(2) 0.795

(3) 1

(4) 3.18

67. The volume of air required for complete combustion of 20 mL of ethene is

(1) 300 mL

(2) 60 mL

(3) 200 mL

(4) 350 mL

68. A quantity of 0.22 g of a gas occupies a volume of 112 mL at pressure of 1 atm and temperature of 273K. The gas may be

(1) NO₂

(2) N₂O

(3) CO₂

(4) C₃H₈

69. Haemoglobin contains 0.4% of iron by mass. Minimum molecular mass of haemoglobin is (Given : Atomic mass of Fe is 56)

(1) 140000

(2) 1400

(3) 14000

(4) 140

70. 10 mL gaseous hydrocarbon on combustion gives 40 mL of CO₂(g) and 50 mL of water vapour. Total number of carbon and hydrogen atoms in the hydrocarbon is _____.

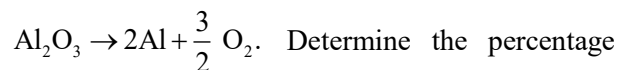
(1) 14

(2) 20

(3) 18

(4) 12

71. 170 g of sample of Al₂O₃ produce 18 g Al. According to following equation



Determine the percentage purity of Al₂O₃ in the original sample.

(1) 2%

(2) 10%

(3) 30%

(4) 20%

72. How many moles of potassium chlorate to be heated to produce 5.6 litre oxygen at STP?

- (1) $\frac{1}{6}$ mol (2) 6 mol
(3) 18 mol (4) $\frac{1}{18}$ mol

73. The quantum numbers of four electrons are given below

I. $n = 4; \ell = -2; s = -\frac{1}{2}$

II. $n = 3; \ell = 1; s = +\frac{1}{2}$

III. $n = 4; \ell = 0; s = +\frac{1}{2}$

IV. $n = 3; \ell = -1; s = +\frac{1}{2}$

The correct decreasing order of energy of these electrons is

- (1) IV > II > III > I
(2) I > III > II > IV
(3) III > I > II > IV
(4) I > II > III > IV

74. The energy of an electron in the ground state ($n = 1$) for He^+ ion is $-x$ J, then that for an electron in ($n = 2$) state for Be^{3+} ion in J is

- (1) $-x$ (2) $-\frac{x}{9}$
(3) $-4x$ (4) $-\frac{4}{9}x$

75. The uncertainty in the position of an electron ($m = 9.1 \times 10^{-28}$ g) moving with a velocity of $3 \times 10^4 \text{ cm s}^{-1}$ accurate upto 0.011% will be:

- (1) 1.92 cm (2) 7.68 cm
(3) 0.175 cm (4) 3.84 cm

76. If the speed of electron in Bohr's first orbit of hydrogen atom be x , then speed of the electron in 3rd orbit is :

- (1) $\frac{x}{9}$ (2) $\frac{x}{3}$ (3) $3x$ (4) $9x$

77. Which of the following transition have minimum wavelength-

- (1) $n_4 \longrightarrow n_1$ (2) $n_2 \longrightarrow n_1$
(3) $n_4 \longrightarrow n_2$ (4) $n_3 \longrightarrow n_1$

78. Match the following columns-

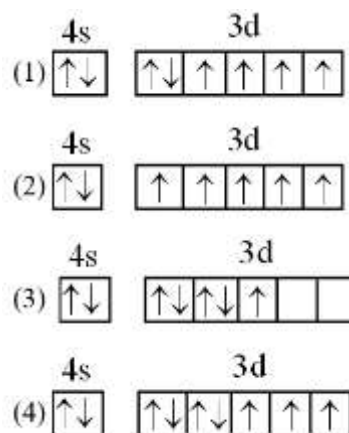
Column-I		Column-II	
A.	$mvr = \frac{nh}{2\pi}$	(i)	Paschen series
B.	Infrared	(ii)	Electron's total energy
C.	$\lambda = h/p$	(iii)	de-Broglie's equation
D.	$\frac{-KZe^2}{2r}$	(iv)	Schrodinger equation
E.	$\hat{H}\Psi = E\Psi$	(v)	Bohr's equation

- (1) A-(v), B-(ii), C-(iii), D-(i), E-(iv)
(2) A-(v), B-(i), C-(iii), D-(ii), E-(iv)
(3) A-(iii), B-(ii), C-(iv), D-(iv), E-(i)
(4) A-(iv), B-(i), C-(ii), D-(iii), E-(v)

79. $\Psi^2 = 0$ represents-

- (1) Node
(2) Orbital
(3) Angular wave function
(4) Wave function

80. The orbital diagrams for Mn is correctly represented as :



81. A green bulb and a red bulb are emitting the radiations with equal power. The correct relation between numbers of photons emitted by bulbs per second is -

- (1) $n_g = n_r$
(2) $n_g > n_r$
(3) $n_g < n_r$
(4) unpredictable

82. A photosensitive surface is receiving light of wavelength $5000 \text{ \AA}$ at the rate of 10^{-7} J/s the number of photon received per second is-
- (1) 2.5×10^{11} (2) 2.5×10^{32}
(3) 2.5×10^{18} (4) 2.5×10^9
83. Suppose the mass of electron decreased by 25 %. How will it affect the Rydberg's constant?
- (1) It remains unchanged
(2) It becomes one fourth
(3) It reduces to 75% of its original value
(4) It is doubles
84. Which of the following observations was not correct during Rutherford's scattering experiment
- (1) Most of the α -particles passed through the gold foil undeflected
(2) A small fraction of the α -particles was deflected by small angles.
(3) A large number of the α -particles were bounce back.
(4) A very few α -particles (1 in 20,000) were bounce back.
85. Orbital having 3 angular nodes and 3 total nodes is
- (1) 5p (2) 3d
(3) 4f (4) 6d
86. The number of d-electrons in Fe^{2+} ($Z=26$) is not equal to the number of electrons in which one of the following:
- (1) p - electrons in Ne ($Z=10$)
(2) s - electrons in Mg ($Z=12$)
(3) p - electrons in Cl ($Z=17$)
(4) d - electrons in Fe ($Z=26$)
87. Ultraviolet light of 6.2 eV falls on an aluminum surface. The threshold energy for a metal is 4.2 eV. The kinetic energy (in Joule) of an electron emitted is
- (1) 3.2×10^{-21} (2) 3.2×10^{-17}
(3) 3.2×10^{-19} (4) 3.2×10^{-15}
88. **Assertion:** The characteristics of cathode rays do not depend upon the material of electrodes and the nature of the gas present in the cathode ray tube.
Reason: Cathode rays consist of negatively charged particles called electrons.
- (1) Both assertion and reason are true and reason is the correct explanation of assertion.
(2) Both assertion and reason are true but reason is not the correct explanation of assertion.
(3) Assertion is true but reason is false.
(4) Both assertion and reason are false.
89. **Assertion:** According to de Broglie, the wavelengths associated with electrons and other subatomic particles can be detected experimentally.
Reason: The wavelength associated with any material particle is directly proportional to its mass.
- (1) Both assertion and reason are true and reason is the correct explanation of assertion.
(2) Both assertion and reason are true but reason is not the correct explanation of assertion.
(3) Assertion is true but reason is false.
(4) Both assertion and reason are false.
90. **STATEMENT-I:** One mole of SO_2 contains double the number of molecules present in one mole of O_2 .
STATEMENT-II: Molecular weight of SO_2 is double to that of O_2 .
- (1) Both Statement I and Statement II are correct
(2) Both Statement I and Statement II are incorrect
(3) Statement I is correct but Statement II is incorrect
(4) Statement I is incorrect but Statement II is correct

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91. Match the columns and select the **correct** option:

	Column-I		Column-II
(a)	Red algae	(i)	<i>Sargassum</i>
(b)	Brown algae	(ii)	<i>Chara</i>
(c)	Green algae	(iii)	<i>Gracilaria</i>

- (1) a-(i), b-(iii) and c-(ii)
 (2) a-(iii), b-(i) and c-(ii)
 (3) a-(iii), b-(ii) and c-(i)
 (4) a-(ii), b-(iii) and c-(i)

92. In pteridophytes :-

- (a) Spores are always of two kinds.
 (b) Vascular tissues are present.
 (c) The sporophyte is parasitic over the photosynthetic gametophyte
 (d) The development of zygotes into young embryos takes place within the female gametophytes.
 (1) (a), (b) and (d) are correct.
 (2) Only (d) is incorrect.
 (3) All are correct except (b).
 (4) (a) and (c) are incorrect.

93. Which of the following is filamentous algae?

- (1) *Fucus* (2) *Chlamydomonas*
 (3) *Spirogyra* (4) *Volvox*

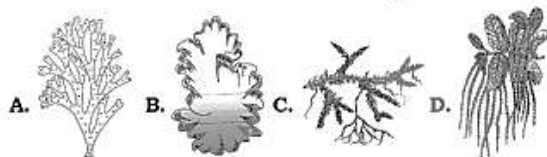
94. *Petromyzon* is not a true fish because of :-

- (1) Absence of operculum and air bladder
 (2) Absence of paired fins and scales
 (3) Absence of cranium and vertebral column
 (4) Absence of gills and median fins

95. Select the mis-match:

- (1) *Rana* - external ear opening
 (2) *Pavo* - Feathers
 (3) *Naja* - Epidermal scales
 (4) *Aplysia* - Radula organ

96. Examine the figures A, B, C & D select the option in which their names are **correctly** mentioned :



	A	B	C	D
(1)	<i>Dictyota</i>	<i>Porphyra</i>	<i>Salvinia</i>	<i>Selaginella</i>
(2)	<i>Porphyra</i>	<i>Dictyota</i>	<i>Selaginella</i>	<i>Salvinia</i>
(3)	<i>Porphyra</i>	<i>Laminaria</i>	<i>Lycopodium</i>	<i>Equisetum</i>
(4)	<i>Dictyota</i>	<i>Porphyra</i>	<i>Selaginella</i>	<i>Salvinia</i>

97. Radial symmetry is not found in animals of phylum

- (1) Ctenophora (2) Hemichordata
 (3) Coelenterata (4) Echinodermata

98. Match the following column-I and column-II.

	Column-I		Column-II
a	Binomial nomenclature	(i)	Carolus Linnaeus
b	Generic name	(ii)	Muscidae
c	Family	(iii)	<i>Panthera</i>
d	Systema Naturae		

- (1) a →(i); b →(iii); c →(ii); d →(ii)
 (2) a →(i); b →(iii); c →(ii); d →(i)
 (3) a →(ii); b →(i); c →(i); d →(iii)
 (4) a →(iii); b →(i); c →(ii); d →(i)

99. Which of the following animal does not belong to division Gnathostomata ?

- (1) Hag fish (2) Dog fish
 (3) *Labeo* (4) *Pristis*

100. Match the following column-I and column-II.

	Column-I		Column-II
a	Rust of wheat	(i)	Zygospor
b	Puffballs	(ii)	<i>Puccinia</i>
c	<i>Neurospora</i>	(iii)	Basidiocarp
d	<i>Albugo</i>	(iv)	Used extensively in biochemical and genetic work

- (1) a →(i); b →(iii); c →(iv); d →(ii)
 (2) a →(iii); b →(ii); c →(iv); d →(i)
 (3) a →(ii); b →(i); c →(iv); d →(iii)
 (4) a →(ii); b →(iii); c →(iv); d →(i)

101. Which fish is kept in aquarium?

- (1) *Betta* (2) *Exocoetus*
 (3) *Pterophyllum* (4) Both (1) and (3).

102. Which of the following fish has gills without operculum:

- (1) *Betta* (Fighting fish)
 (2) *Clarius* (Magur)
 (3) *Charcharodon* (Largest white shark)
 (4) *Exocoetus* (Flying Fish)

103. Match List I with List II.

	List I (Animal)		List II (Excretory organ)
a.	<i>Taenia</i>	(i)	Nephridia
b.	<i>Sycon</i>	(ii)	Body surface
c.	<i>Periplaneta</i>	(iii)	Flame cells
d.	<i>Pheretima</i>	(iv)	Malpighian tubules

Choose the correct answer from the options given below:

- (1) a-i, b-ii, c-iii, d-iv
 (2) a-i, b-ii, c-iv, d-iii
 (3) a-iii, b-ii, c-iv, d-i
 (4) a-ii, b-i, c-iv, d-iii

104. Which character is similar in *Exocoetus* and *Scoliodon*?

- (1) They have bony vertebral column
- (2) They are marine fishes
- (3) They regulate buoyancy
- (4) They have placoid scales.

Direction: (Q.105 to Q.107) Given below are two statements, Choose the correct answer from the options given below:

- (A) Both statement I and statement II are correct.
- (B) Both statement I and statement II are incorrect.
- (C) Statement I is correct and statement II is incorrect.
- (D) Statement I is incorrect and statement II is correct.

105. **Statement I:** Birds and mammals have a constant body temperature.

Statement II: Cold blooded animals do not regulate body temperature.

- (1) A
- (2) B
- (3) C
- (4) D

106. **Statement I:** The Skeleton of most of sponges is made up of spicules

Statement II: Bath sponges don't have spicules and therefore used for rubbing skin.

- (1) A
- (2) B
- (3) C
- (4) D

107. **Statement I:** All coelenterates have cnidocytes for offence and defence.

Statement II: Alternation of *polyp* & medusa forms is exhibited by *Obelia*.

- (1) A
- (2) B
- (3) C
- (4) D

108. The following features belongs to which phylum?

- (a) Exclusively marine
- (b) Radial symmetry
- (c) Triploblastic
- (d) organ system level organization
- (1) Coelentrata
- (2) Ctenophora
- (3) Echinodermata
- (4) Platyhelminthes

109. The unique mammalian characteristics are:

- (1) Body hair, tympanic membrane and mammary glands
- (2) Body hair, pinna and mammary glands
- (3) Body hair, pinna and indirect development
- (4) Pinna, monocondylic skull and mammary glands

110. Which of the following animals are true Coelomates with bilateral symmetry?

- (1) Adult Echinoderms
- (2) Aschelminthes.
- (3) Platyhelminthes.
- (4) Annelids.

111. Match the animal with its character & phylum/ class

Column I	Column II	Column III
(1) <i>Ichthyophis</i>	Limbless	Amphibia
(2) <i>Myxine</i>	Ectoparasite	Chondrichthyes
(3) <i>Limulus</i>	Terrestrial	Mollusca
(4) <i>Adamsia</i>	Radial symmetry	Porifera

112. The choanocyte cells are the characteristic feature of Porifera. It is related with all the following except one.

- (1) Maintains flow of water current
- (2) Lines spongocoel and canals
- (3) Intracellular digestion
- (4) Repair and regeneration

113. Which of the following is a true fish, a vertebrate animal.

- (1) Cuttle fish
- (2) Star fish
- (3) Jelly fish
- (4) Fighting fish

114. If air bladder gets damaged in bony fishes then –

- (1) Bony fishes will continue to float in the water
- (2) It does not affect the movement of bony fishes.
- (3) They may sink in the water
- (4) Bony fishes don't have air bladder.

115. Find out the incorrect matching:

- (1) Reptiles– *Chelone*, *Testudo*, *Chameleon*
- (2) Aves – *Psittacula*, *Aptenodytes*, *Neophron*
- (3) Mammals– *Elephas*, *Rattus*, *Delphinus*
- (4) Amphibian– *Bufo*, *Rana*, *Calotes*

116. Which of the following organism has complete alimentary canal with well developed muscular pharynx ?

- (1) Earthworm
- (2) Flatworm
- (3) Round worm
- (4) Tape worm

117. Match the list of animals with their level of organization and choose the correct sequence. :-

	Column-I		Column-II
a.	Organ level	(p)	<i>Pheretima</i>
b.	Cellular level	(q)	<i>Fasciola</i>
c.	Tissue level	(r)	<i>Spongilla</i>
d.	Organ system level	(s)	<i>Obelia</i>

- (1) a–(s), b–(r), c–(p), d–(q)
- (2) a–(s), b–(q), c–(r), d–(p)
- (3) a–(q), b–(s), c–(r), d–(p)
- (4) a–(q), b–(r), c–(s), d–(p)

118. In birds:-

- (1) Oesophagus is modified into Crop for quick food ingestion and storage and Gizzard for crushing the food.
- (2) Scales are found only on hind limbs.
- (3) Air sacs are also found attached to lungs
- (4) Possess all the features mentioned above

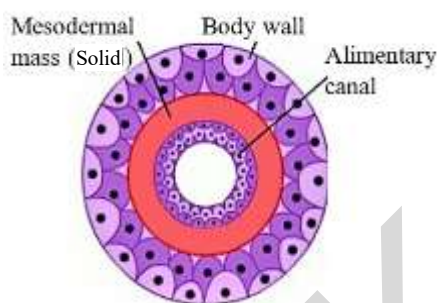
119. Which of the following is incorrect for the class Chondrichthyes?

- (1) They have 3 chambered heart
- (2) Some of them like *Torpedo* have electric organs
- (3) Some of them like *Trygon* possess poison sting
- (4) Are cold blooded animals

120. Choose the odd one out w.r.t. *Petromyzon*:

- (1) Cranium and vertebral column are cartilaginous
- (2) Jaws, scales and paired fins present.
- (3) After spawning, within few days, they die
- (4) Their larvae, after metamorphosis, return to the ocean

121. The cross-section of the body of an acoelomate animal is given below. Identify the animal.



- (1) Cockroach
- (2) Roundworm
- (3) *Planaria*
- (4) Earthworm

122. In some animal groups, the body is found divided into compartments with at least some organs repeated. This characteristic feature is named :

- (1) Metachrosis
- (2) Metamerism
- (3) Metagenesis
- (4) Metamorphosis

123. *Rana* and *Hyla* are the members of class-Amphibia bearing which of the following characters?

- (1) A tympanum represents the ear.
- (2) They need water for breeding
- (3) They have moist skin without scales.
- (4) All of these

124. Ventral, solid and double nerve cord is characteristic features of -

- (1) Chordata
- (2) Arthropoda
- (3) Urochordata
- (4) Vertebrata

125. Match the following genera with their respective phylum.

Genus		Phylum	
a	<i>Ophiura</i>	(i)	Mollusca
b	<i>Physalia</i>	(ii)	Platyhelminthes
c	<i>Pinctada</i>	(iii)	Echinodermata
d	<i>Planaria</i>	(iv)	Coelenterata

Select the correct option :

- (1) a-iv b-i c-iii d-ii
- (2) a-iii b-iv c-i d-ii
- (3) a-i b-iii c-iv d-ii
- (4) a-ii b-iv c-iii d-i

126. Identify the incorrect statements about the class Chondrichthyes.

- a. They are exclusively freshwater organisms.
- b. The skin is covered with placoid scales.
- c. The heart is a two-chambered.
- d. Some cartilaginous fishes, like electric rays, have electric organs.

- (1) a, d
- (2) b and c
- (3) b, c, d
- (4) a, b, c, d

127. Choose the correct match.

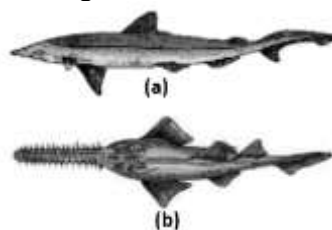
- (1) *Psittacula* and *Neophron* – Air sacs connected to lungs to supplement respiration
- (2) *Bufo* and *Salamandra* – Ear pinna represents ear.
- (3) *Chelone* and *Testudo* – external fertilization and direct development
- (4) *Physalia* and *Ctenoplanea*– Cnidoblasts are absent

**128. (i) Circulatory system is of open type
(ii) Eyes may be compound or simple
(iii) Are mostly monoecious
(iv) Are mostly viviparous**

Which of the above statements are **correct** regarding **Arthropods** ?

- (1) (ii), (iii), (iv)
- (2) (i), (ii), (v)
- (3) (iii), (iv), (v)
- (4) (i), (ii), (iii)

129. Identify the organism and their common name :



- (1) a-Hippocampus (sea horse); b-Catla (Katla)
- (2) a-Clarias (Magur); b-Labeo (Rohu)
- (3) a-Labeo (Rohu); b-Clarias (Magur)
- (4) a-Scoliodon (Dog fish); b-Pristis (Saw fish)

130. Match column-I with Column-II.

	Column-I		Column-II
(i)	High power of regeneration	(p)	<i>Ascaris</i>
(ii)	Radial symmetric and Coelomate	(q)	Sea anemone
(iii)	Bilateral symmetric and Pseudocoelomate	(r)	Starfish
(iv)	Gastrovascular cavity and hypostome.	(s)	<i>Planaria</i>

- (1) (i)-s, (ii)-r, (iii)-p (iv)-q
 (2) (i)-p (ii)-q (iii)-r, (iv)-s
 (3) (i)-q, (ii)-p, (iii)-r (iv)-s
 (4) (i)-r (ii)-s (iii)-p, (iv)-q

131. Read the following statements and mark the incorrect statement: –

- (1) Duck billed platypus is a connecting link between reptiles and mammals
 (2) Endoskeleton of birds consist of hollow bones.
 (3) The unique features of mammals are the presence of mammary glands
 (4) All the mammals have biconcave and enucleated RBCs

132. Which of the following is not a feature of *Ornithorhynchus* commonly called 'Duck billed platypus'?

- (1) Oviparity
 (2) Mammary glands without nipples.
 (3) Body hair and pinna.
 (4) Feathers on body

133. The animal shown in the adjacent figure is characterised by :



- (1) Choanocytes
 (2) Water vascular system
 (3) Radial symmetry
 (4) External fertilization

134. (i) Recent position is non-chordates.
 (ii) Possess metameric segmentation
 (iii) Earlier considered as chordates
 (iv) Respiration occur through gills
 (v) Sexes separate and external fertilisation
 Which of the above statements are **correct** regarding **hemichordates**?

- (1) (iii), (iv), (v) only
 (2) (i), (ii) only
 (3) (i), (ii), (iii) only
 (4) (i), (iii), (iv), (v) only

135. Which of following statement is/are incorrect

- (1) Medusae form polyps sexually and polyps produce medusae asexually
 (2) Sponges are hermaphrodite and show internal fertilisation
 (3) Bilateral symmetrical animal can divided into two identical halves by only one plane
 (4) Platyhelminthes has complete alimentary canal with mouth at anterior end while anus at posterior end.

136. **Statement I:** Chemosynthetic autotrophic bacteria oxidise various inorganic substances such as nitrates, nitrites and ammonia, and use the released energy for ATP production.

Statement II: These chemosynthetic autotrophs play an important role in recycling nutrients like nitrogen, phosphorus, iron and sulphur, but their activity makes soil less fertile.

Which of the following is correct?

- (1) Both statements I and II are correct
 (2) Statement I is correct but Statement II is incorrect
 (3) Statement I is incorrect but Statement II is correct
 (4) Both statements I and II are incorrect.

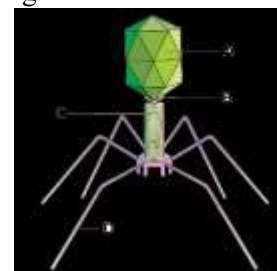
137. **Statements-1:** Yeast is a multicellular fungus.

Statements-2: *Penicillium* is unicellular fungus.

Statements-3: *Albugo* is a parasitic fungus on mustard.

- (1) Only statement-1 and statement-2 are correct
 (2) All the above statements are incorrect
 (3) Only statement-3 is correct
 (4) Both statement-1 and statement-3 are correct

138. Identify A, B, C, and D parts in this diagram of bacteriophage.



- (1) A – Head, B – Sheath, C – Collar, D – Tail fibres
 (2) A – Head, B – Collar, C – Sheath, D – Tail fibres
 (3) A – Head, B – Collar, C – Tail fibres, D – Sheath
 (4) A – Head, B – sheath, C – Tail fibres, D – Collar

139. There exists a close association between the alga and the fungus called lichen, The fungus
- (1) Provides protection, anchorage and Absorption of minerals for the algae
 - (2) Provides food for the alga
 - (3) Fixes the atmospheric nitrogen for the alga
 - (4) Releases oxygen for the alga

140. Kelps are massive
- (1) Brown algae
 - (2) Amphibious plants
 - (3) Flowering plants
 - (4) Plants with naked seeds

141. Which of the following lack heterospory ?
- (1) *Dryopteris*
 - (2) *Selaginella*
 - (3) *Salvinia*
 - (4) *Cycas*

142. Non-motile gametes are characteristically found in
- (1) *Ulothrix*
 - (2) *Porphyra*
 - (3) *Laminaria*
 - (4) *Volvox*

143. Read the following statements and select the **incorrect** ones :
- (a) Mosses have an elaborate mechanism of spore dispersal.
 - (b) In liverworts, the haploid free living sporophyte is formed by spore germination.
 - (c) Vegetative reproduction in *Polytrichum* occurs by budding in the secondary protonema.
 - (d) *Marchantia* is a bryophyte.
 - (e) Growth of bog moss ultimately fill up ponds and lakes with soil.
- (1) (c) only
 - (2) (d) only
 - (3) (b) only
 - (4) All except (a)

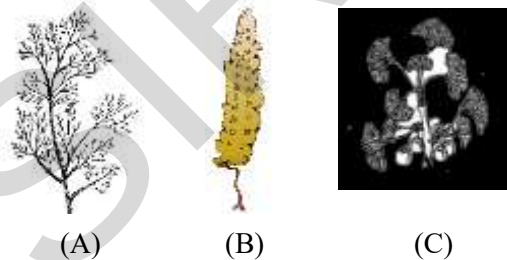
144. Select the correct statement.
- (1) Cholera, typhoid, tetanus are well-known diseases caused by viruses
 - (2) Dinoflagellates, euglenoids and slime moulds are placed under kingdom Monera
 - (3) Members of kingdom Protista are primarily aquatic
 - (4) Dinoflagellates are the chief 'producers' in the oceans

145.are saprophytic protists, whose body moves along decaying twigs and leaves engulfing organic material.
- (1) Euglenoids
 - (2) Dinoflagellates
 - (3) Chrysophytes
 - (4) Slime moulds

146. **Statement** :- Euglenoids are primarily heterotrophic but can behave like autotrophs in the absence of light.

Which of the following is **correct** ?

- (1) The statement is true, they are mainly heterotrophs but switch to autotrophy in darkness
 - (2) The statement is false, because euglenoids are strictly heterotrophic and never photosynthetic
 - (3) The statement is false, because euglenoids are primarily autotrophic but can behave like heterotrophs in the absence of sunlight
 - (4) The statement is false, because euglenoids possess a chitinous wall that prevents photosynthesis
147. Examine the figure A, B and C, In which one of the four options all the item A, B and C are correct?



	(A)	(B)	(C)
(1)	<i>Fucus</i>	<i>Selaginella</i>	<i>Ginkgo</i>
(2)	<i>Polysiphonia</i>	<i>Ectocarpus</i>	<i>Ginkgo</i>
(3)	<i>Polysiphora</i>	<i>Ectocarpus</i>	<i>Salvinia</i>
(4)	<i>Fucus</i>	<i>Selaginella</i>	<i>Salvinia</i>

Direction (Q.148 to 151) Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R** Choose the correct answer from the options given below-

- (A) Both A and R are true and R is the correct explanation of A
- (B) Both A and R are true but R is not the correct explanation of A.
- (C) A is true but R is false
- (D) A is false but R is true.

148. **Assertion:** In gymnosperms, the pollen grains are released from the microsporangium and carried by air currents.

Reason: Air currents carry the pollen grains to the mouth of archegonia where the male gametes are discharged and pollen tube is not formed.

- (1) A
- (2) B
- (3) C
- (4) D

149. Assertion : The first stage of gametophyte in the life cycle of moss is protonema stage.

Reason: Protonema develops directly from spores produced in capsule.

- (1) A (2) B (3) C (4) D

150. Assertion : Angiosperms have reduced male gametophyte.

Reason: In Angiosperms male gametophyte has two or three cells.

- (1) A (2) B (3) C (4) D

151. Assertion : Bryophytes possess archegonium as a female sex organ

Reason: Algae also possess the archegonium.

- (1) A (2) B (3) C (4) D

152. Which of the following statements about the vegetative organs of gymnosperms is correct?

- (1) The roots of *Cycas* show coralloid structures that are associated with nitrogen fixing cyanobacteria.
(2) The stem of gymnosperms is always unbranched, as in *Pinus* and *Cedrus*.
(3) The leaves of *Cycas* are needle like, whereas those of *Pinus* are pinnate and large.
(4) Gymnosperm leaves are generally short lived and herbaceous, like in angiosperms.

153. Stored food of brown algae is :-

- (1) Glycogen
(2) Starch
(3) Laminarin and mannitol
(4) Floridean starch

154. Choose the correct statements

- A. Phylum Chordata is divided into three subphyla: Urochordata, Cephalochordata and Vertebrata.
B. Cyclostomes have a sucking and circular mouth with jaws.
C. Due to the absence of air bladder in Chondrichthyes, they have to swim constantly to avoid sinking.
D. Osteichthyes have four pairs of gills which are covered by an operculum on each side.

- (1) A, B and C (2) A, C and D
(3) B, C and D (4) A, B, C and D

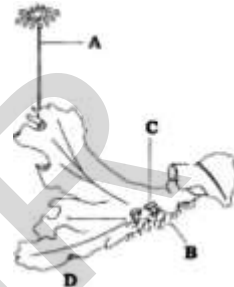
155. Protists obtain their food by :

- (a) Chemosynthesis
(b) Photosynthesis
(c) Heterotrophic mode of nutrition
(1) (a) only (2) (a) and (c)
(3) (a) and (b) (4) (b) and (c)

156. Peat(fossil fuel) is obtained from which bryophyte?

- (1) *Sphagnum* (2) *Funaria*
(3) *Riccia* (4) *Porella*

157. Select correct option for marking (A), (B), (C), (D) and its feature :



- (1) A-Archegoniophore which bear antheridium.
(2) C-Small, cup shaped receptacles in which sexual spores are formed.
(3) B-helps in attachment with substratum.
(4) D-Include in a plant group that have elaborate mechanism of spore dispersal.

158. Alternation of generation or life cycle of fungi specially sac fungi completed by following steps:

- (1) Plasmogamy → Karyogamy → Meiosis
(2) Plasmogamy → Dikaryon phase → Karyogamy → Meiosis
(3) Meiosis → Karyogamy → Plasmogamy
(4) Karyogamy → Dikaryon phase → Plasmogamy

Direction: (Q.159 to Q.161) Given below are two statements, Choose the correct answer from the options given below :

- (A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is correct.

159. Statement I: Mosses has great ecological importance.

Statement II: They prevent soil erosion.

- (1) A (2) B (3) C (4) D

160. Statement I: Bacteria are cosmopolitan

Statement II: Flagella of bacteria is made up of tubulin protein

- (1) A (2) B (3) C (4) D

161. Statement I: In gymnosperm haploid endosperm is present.

Statement II: In angiosperm triploid endosperm is present

(1) A (2) B (3) C (4) D

162. Ascospores are produced ____ A ____ in ____ B ____ and basidiospores are ____ C ____ produced on the ____ D ____.

Here A, B, C and D are :-

- (1) A-endogeneously, B-basidium, C-exogenously, D-ascus
- (2) A-endogenously, B-ascus, C-exogenously, D-basidium
- (3) A-exogenously, B-ascus, C-endogenecously, D-basidium
- (4) A-exogenously, B-basidium, C-endogenously, D-ascus

163. The sporophyte in mosses is :

- (1) Less elaborated than that in liverworts
- (2) More elaborated than that in liverworts
- (3) equally elaborated than that in liverworts
- (4) independent of gametophyte

164. If *Equisetum* belongs to sphenopsida, then *Psilotum* and *Pteris* belong to respectively:

- (1) Lycopsida and Psilopsida
- (2) Lycopsida and Pteropsida
- (3) Psilopsida and Lycopsida
- (4) Psilopsida and Pteropsida

165. The photosynthetic protists are -

- (1) Diatoms, euglenoids and slime moulds
- (2) *Amoeba*, dinoflagellates and diatoms
- (3) Euglenoids, diatoms and dinoflagellates
- (4) Ciliated Protozoans, zooflagellates and dinoflagellates

166. Refer to the statement and answer the question.

"Once the sexual stage of members of deuteromycetes were discovered, they were often moved to X and Y."

Identify X and Y.

- (1) X - Monera ; Y - Protista
- (2) X - Basidiomycetes ; Y - Phycomycetes
- (3) X - Ascomycetes ; Y - Basidiomycetes
- (4) X - Phycomycetes ; Y - Archaeobacteria

167. Read the following statements w.r.t. binomial nomenclature system.

Correct statements are:

- A. It was developed by Linnaeus.
- B. Name of the author is written after specific epithet in abbreviated form.
- C. The first word denoting specific epithet starts with a capital letter.
- D. Biological names are generally in Latin and written in italics.

(1) A, B and D (2) A, C and D
(3) B, C and D (4) Only A and B

168. Which of the following are characters of dinoflagellates:

- A. They are saprophytes
- B. They appear yellow, green, brown, blue, and red colour.
- C. The cell wall has stiff cellulose plates on the outer surface
- D. They have two flagella on anterior side one short and another long

(1) C and D (2) A, B and D
(3) B, C and D (4) B and C

169. Mycorrhizal association is responsible for:

- a. Increasing absorptive area
- b. Providing shelter to fungi
- c. Nitrogen fixation
- d. Enhanced supply of N, P, S

(1) Only a and b
(2) Only b and c
(3) Only a, b and d
(4) All a, b, c and d

170. Chemosynthetic autotrophic bacteria obtain energy for their ATP formation from :

- (1) Oxidation of always organic substances.
- (2) Oxidise to various inorganic substances.
- (3) Oxidise to nitrite into nitrate
- (4) Both (2) & (3)

171. Select the wrong pair :-

- (1) Carrageen - Red algae
- (2) Algin - Brown algae
- (3) Agar-Agar - *Gelidium*
- (4) Floridian starch - Shows similarity with chitin.

172. All are edible fungus except :-

- (1) *Morels* (2) *Agaricus*
(3) *Truffles* (4) *Amannita*

173.

Kingdom		Organism
(i)	Monera	Archaeobacteria, Eubacteria, BGA, Euglenoids
(ii)	Protista	Slime moulds, protozoans, Euglenoids
(iii)	Fungi	<i>Albugo</i> , <i>Mucor</i> , <i>Penicillium</i> , Slime moulds

Which is/are the correct match (es)?

- (1) Only (i) (2) Only (ii)
(3) Only (iii) (4) All are correct

174. A fungus that is decomposer of wood, has long lived secondary mycelium, shows clamp connections and produces exogenous sexual spores is put under the class:

- (1) Zygomycetes
(2) Basidiomycetes
(3) Ascomycetes
(4) Oomycetes

175. Heterocysts are specialised cells found in some BGA. They are specialised for :

- (1) Sexual reproduction
(2) Nitrogen fixation under anaerobic condition
(3) Nitrogen fixation under aerobic condition
(4) Performing photosynthesis with the help of PS II

176. As compared to slime moulds, euglenoids have:

- (A) Chloroplast
(B) Holophytic nutrition
(C) Proteinaceous pellicle
(D) Contractile vacuole

- (1) A and B (2) B and C
(3) A, B and C (4) A, B, C and D

177. Match the columns, and choose the correct combination from the options given below.

Column-I (Disease)		Column-II (Pathogen)	
(A)	Bovine spongiform encephalopathy (BSE)	(i)	Virus
(B)	Mad cow disease	(ii)	Viroid
(C)	Potato spindle tuber disease	(iii)	Prion
(D)	Mumps	(iv)	Bacteria

- (1) A-(iv), B-(i), C-(ii), D-(iii)
(2) A-(iii), B-(ii), C-(i), D-(iv)
(3) A-(iii), B-(iii), C-(ii), D-(i)
(4) A-(iv), B-(iii), C-(ii), D-(i)

178. Lichens cannot tolerate ____ (A) ____ especially due to ____ (B) ____.

Fill the blanks with suitable option for (A) and (B).

(A)	(B)
(1) Water pollution	CO ₂
(2) Soil pollution	CH ₄
(3) Air pollution	SO ₂
(4) Air pollution	CO ₂

179. Meiosis occurs in pteridophytes:

- (1) During gamete formation
(2) During spore formation
(3) During the development of oospore
(4) None of these

180. Cytotaxonomy and chemotaxonomy are based on respectively -

- (1) Chromosome number and chromosome structure.
(2) Chromosome structure and chromosome behaviour.
(3) Chromosome number and chemical constituent
(4) Chemical constituents and chromosome number.

Answer Key and Solution

ANSWER KEY & SOLUTION

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	4	3	2	1	4	1	4	4	2	4	1	4	3	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	4	2	4	4	3	2	4	4	2	1	4	3	2	2	2
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	3	4	3	1	1	3	1	3	4	2	1	3	1	2	3
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	3	4	1	4	3	2	4	3	1	3	4	2	3	4	1
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	3	3	2	1	2	3	1	3	3	1	4	1	2	1	3
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	2	1	2	1	2	3	1	3	3	3	3	3	2	3	4
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Ans.	2	4	3	2	1	4	2	2	1	4	4	3	3	2	1
Que.	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Ans.	1	1	3	2	4	1	4	4	3	4	3	4	4	1	2
Que.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Ans.	3	2	4	2	2	3	1	2	4	1	4	4	1	4	4
Que.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
Ans.	2	3	2	1	1	1	2	3	3	4	3	2	3	1	1
Que.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
Ans.	3	1	3	2	4	1	3	2	1	3	1	2	2	4	3
Que.	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
Ans.	3	1	4	3	4	4	4	2	2	2	4	3	3	2	3

PHYSICS

Q.1 (3)

Sol. Here $\sqrt{1 + \frac{2k\ell}{ma}}$ is a number.

It's a dimensionless quantity.

$$\left[\frac{2k\ell}{ma} \right] = [M^0 L^0 T^0]$$

$$[K] = \frac{[m][a]}{[\ell]}$$

$$= \frac{M^1 L^1 T^{-2}}{[L]} = M^1 L^0 T^{-2}$$

So dimension of [b] is

$$[b] = \left[\frac{ma}{K} \right] = \left[\frac{MLT^{-2}}{MT^{-2}} \right]$$

$$[b] = L$$

unit of b is metre

Q.2 (4)

Sol. $[\alpha] = \frac{[F]}{v^2} = \frac{[M^1 L^1 T^{-2}]}{[L^1 T^{-1}]^2} = M^1 L^{-1} T^0$

$$[\beta][t] = 1$$

$$[\beta] = [T^{-1}]$$

Q.3 (3)

Sol. $\int \sin x \, dx + 2 \int e^x \, dx + 4 \int \frac{1}{x} \, dx$
 $= -\cos x + 2e^x + 4 \log_e x + c$

Q.4 (2)

Sol. $v \propto \lambda \rho g$

$$v = k\lambda^a \rho^b g^c$$

$$[M^0 L^1 T^{-1}] = K[M^0 L^1 T^0]^a [M^1 L^{-3} T^0]^b [M^0 L^1 T^{-2}]^c$$

$$[M^0 L^1 T^{-1}] = K[M^b L^{a-3b+c} T^{-2c}]$$

Comparing both sides

$$b = 0$$

$$-1 = -2c \Rightarrow c = \frac{1}{2}$$

$$1 = a - 3b + c$$

$$1 = a - 0 + \frac{1}{2}$$

$$a = \frac{1}{2}, V = K\lambda^{1/2} \rho^0 g^{1/2}$$

squaring both sides

$$V^2 = kg\lambda$$

Q.5 (1)

Sol. $P = \frac{W}{t} = F \times \frac{s}{t} = \frac{\text{mass} \times \text{acc.} \times \text{distance}}{t}$

$$1 \text{ unit of } P = \frac{20 \times \frac{10}{5^2} \times 10}{5} = 16W$$

Q.6 (4)

Sol. $\frac{\Delta x}{x} = 1\%$

$$\frac{\Delta y}{y} = 3\%$$

$$\frac{\Delta z}{z} = 2\%$$

$$t = \frac{xy^2}{z^3}$$

$$\left(\frac{\Delta t}{t} \right) \times 100 = \left(\frac{\Delta x}{x} + \frac{2\Delta y}{y} + \frac{3\Delta z}{z} \right) \times 100$$

$$\frac{\Delta t}{t} \times 100 = 1 + 2(3) + 3(2)$$

$$= 1 + 6 + 6 = 13\%$$

Q.7 (1)

Sol. $\bar{x} = \frac{90 + 91 + 95 + 92}{4} = 92$

$$\Delta x_1 = 90 - 92 = -2$$

$$\Delta x_2 = 91 - 92 = -1$$

$$\Delta x_3 = 95 - 92 = 3$$

$$\Delta x_4 = 92 - 92 = 0$$

$$\overline{\Delta x} = \frac{\Delta x_1 + \Delta x_2 + \Delta x_3 + \Delta x_4}{4}$$

$$|\overline{\Delta x}| = \frac{2 + 1 + 3 + 0}{4} = 1.5$$

$$\text{Time} = (92 \pm 1.5)$$

Q.8 (4)

Sol. Key idea horizontal ranges are same for complementary angles of projection i.e., for θ and $(90^\circ - \theta)$

We know that if two stones have same horizontal range, then this implies that both are projected at θ and $90^\circ - \theta$

$$\text{Here, } \theta = \frac{\pi}{3} = 60^\circ$$

$$\therefore 90^\circ - \theta = 90^\circ - 60^\circ = 30^\circ$$

For first stone, Maximum height

$$= 102 = \frac{u^2 \sin^2 60^\circ}{2g}$$

For second stone, Maximum height,

$$h = \frac{u^2 \sin^2 30^\circ}{2g}$$

$$\therefore \frac{h}{102} = \frac{\sin^2 30^\circ}{\sin^2 60^\circ} = \frac{\left(\frac{1}{2}\right)^2}{\left(\frac{\sqrt{3}}{2}\right)^2}$$

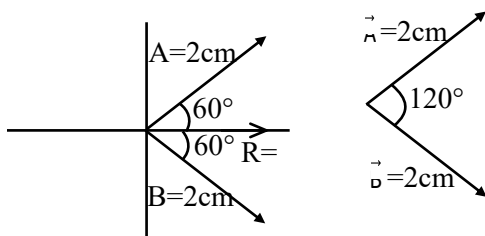
$$\text{Or } h = 102 \times \frac{\frac{1}{4}}{\frac{3}{4}} = 34\text{m}$$

Q.9 (4)

Sol. $\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$

Q.10 (2)

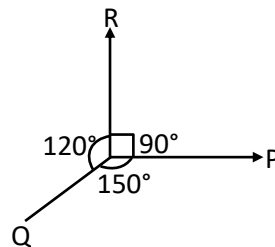
Sol.



Resulting of $\vec{A}$ & $\vec{B}$ also have same unit 2 cm.

Q.11

Sol. (4)



$$\frac{P}{\sin 120^\circ} = \frac{Q}{\sin 90^\circ} = \frac{R}{\sin 150^\circ}$$

$$\frac{P}{\sqrt{3}/2} = \frac{Q}{1} = \frac{R}{\frac{1}{2}}$$

$$P : Q : R = \sqrt{3} : 2 : 1$$

Q.12 (1)

Sol. $|\vec{r}_1| - |\vec{r}_2| = 10 \text{ dyne} \Rightarrow \theta = 120^\circ$

$$|\vec{r}| = 2|\vec{r}_1| \cos \theta/2 \Rightarrow 10 \text{ dyne}$$

Q.13 (4)

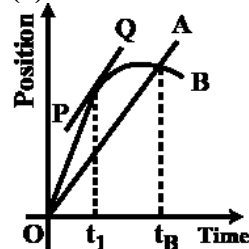
Sol. $|\vec{u}| + |\vec{v}| = |\vec{w}|$

$$1 + 1 + 2 \cdot 1 \cdot 1 \cos \theta = 1$$

$$\cos \theta = -\frac{1}{2}$$

$$\theta \rightarrow 120^\circ$$

Q.14 (3)



Sol.

(3) Both have some velocity at some time before t_B at time t_1

Q.15 (2)

Sol. In inclined plane initial $u = 0$

$$S = \frac{1}{2} at^2 \text{ and } a = g \sin \theta$$

$$\ell = \frac{1}{2} g \sin \theta \times (4)^2 \dots\dots(i)$$

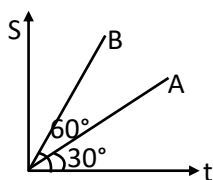
$$\frac{\ell}{4} = \frac{1}{2} g \sin \theta t^2 \dots\dots(ii)$$

From (i) and (ii)

$$t = 2 \text{ sec}$$

Q.16

Sol. (4)



$$\frac{V_A}{V_B} = \frac{\tan 30^\circ}{\tan 60^\circ} \Rightarrow \therefore \frac{V_A}{V_B} = \frac{1/\sqrt{3}}{\sqrt{3}} = \frac{1}{3}$$

Q.17

Sol. (2)

Distance = Total Area = 105 m

Displacement = 90 - 15 = 75 m

Q.18

Sol. (4)

Q.19

Sol. (4)

$$X = \alpha t^2 - \beta t^3$$

$$(1) 0 = \alpha t^2 - \beta t^3 \Rightarrow t = \frac{\alpha}{\beta}$$

$$(2) v = \frac{dx}{dt} = 2\alpha t - 3\beta t^2 \Rightarrow v = 0 \Rightarrow t = \frac{2\alpha}{3\beta}$$

$$(3) a = \frac{d^2x}{dt^2} = 2\alpha - 6\beta t$$

$$\text{When } t = 0 \Rightarrow a = 2\alpha; v = 0$$

Q.20

Sol. (3)

Let total height = H

Time of ascent = T

$$\text{So, } H = uT - \frac{1}{2}gT^2$$

Distance covered by ball in time (T-t) sec.

$$y = u(T-t) - \frac{1}{2}g(T-t)^2$$

So distance covered by ball in last t sec.

$$h = H - y$$

$$= \left[uT - \frac{1}{2}gT^2 \right] - \left[u(T-t) - \frac{1}{2}g(T-t)^2 \right]$$

By solving and putting $T = \frac{u}{g}$, we will get

$$h = \frac{1}{2}gt^2$$

Q.21

Sol. (2)

$$\vec{r} = (3\hat{i} + 4\hat{j})2 + \frac{1}{2}(4\hat{j} + 4\hat{j})4$$

$$= 10\hat{i} + 8\hat{j} + 8\hat{i} + 8\hat{j}$$

$$\vec{r} = 18\hat{i} + 16\hat{j}$$

$$\vec{r} = \vec{r}_2 - \vec{r}_1$$

$$\vec{r}_2 - \vec{r}_1 = 18\hat{i} + 16\hat{j}$$

$$\vec{r}_2 = 18\hat{i} + 20\hat{j}$$

$$|\vec{r}_2| = \sqrt{18^2 + 20^2}$$

Q.22

Sol. (4)

At any time acceleration acting in downward direction g is constant.

At highest point, velocity is only in x direction.

Q.23

Sol. (4)

(A→r), (B→s), (C→r), (D→r)

24. (2)

Sol. As we know, $y = a \sin \theta + b \cos \theta$ can be written as:

$$y = \sqrt{a^2 + b^2} \sin \left[\theta + \tan^{-1} \left(\frac{b}{a} \right) \right]$$

$$y = \sqrt{3^2 + 4^2} \sin \left[x + \tan^{-1} \left(\frac{4}{3} \right) \right]$$

$$y = 5 \sin (x + 53^\circ)$$

and $\sin (x + 53^\circ)$ will have its maximum & minimum value equal to +1 & -1 respectively.

$$\text{Hence, } y_{\max} = 5 \text{ \& } y_{\min} = -5$$

25. (1)

Sol. Velocity,

$$v = \frac{dx}{dt} = \frac{d}{dt} (ae^{-pt} + be^{qt})$$

$$= -pae^{-pt} + qbe^{qt}$$

Acceleration,

$$a = \frac{dv}{dt} = \frac{d}{dt} (-pae^{-pt} + qbe^{qt})$$

$$= p^2ae^{-pt} + q^2be^{qt}$$

Acceleration is positive, so velocity goes on increasing with time.

26. (4)

27. (3)

Sol. Time taken to reach the maximum height,

$$t_1 = \frac{u}{g}$$

If t_2 is the time taken to hit the ground, then

$$-H = ut_2 - \frac{1}{2}gt_2^2$$

But $t_2 = nt_1$ (Given)

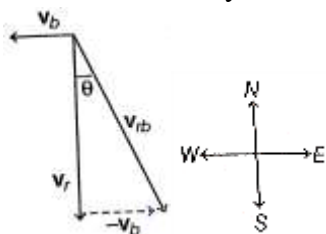
$$\text{So, } -H = u \frac{nu}{g} - \frac{1}{2}g \frac{n^2 u^2}{g^2}$$

$$\Rightarrow -H = \frac{nu^2}{g} - \frac{1}{2} \frac{n^2 u^2}{g}$$

$$\Rightarrow 2gH = nu^2 (n - 2)$$

28. (2)

Sol. In figure below v_r represents the velocity of rain and v_b is the velocity of the bicycle.



Then, angle θ is given by

$$\tan \theta = \frac{v_b}{v_r} = \frac{12}{35} = 0.343$$

$$\text{or } \theta = \tan^{-1}(0.343)$$

So, the cyclist should hold the umbrella to the vertical in the west.

29. (2)

Sol. Given, velocity of girl,

$$v_g = 5\hat{i} \text{ ms}^{-1}$$

Let velocity of rain,

$$v_r = v_x \hat{i} + v_y \hat{j} \text{ ms}^{-1}$$

Relative velocity of rain, $v_{rg} = (v_x - 5)\hat{i} + v_y \hat{j}$

Now, it is vertical,

$$\text{So, } \tan \theta = \frac{v_x - 5}{v_y} = 0$$

$$\Rightarrow v_x = 5 \quad \dots(i)$$

On increasing the speed of the girl, relative velocity becomes $(v_x - 15)\hat{i} + v_y \hat{j}$

$$\tan \theta = \tan 45^\circ = \frac{v_x - 15}{v_y} = 1$$

$$\Rightarrow v_x - 15 = v_y \Rightarrow v_y = -10 \quad [\text{using Eq. (i)}]$$

$\Rightarrow$ Magnitude of velocity of rain

$$= \sqrt{(5)^2 + (10)^2} = 5\sqrt{5} \text{ ms}^{-1}$$

30. (2)

Sol. $v_A = v_x = \frac{dx}{dt} = 8 - 6t$ or v_A at $t = 1$ s,

$$\Rightarrow v_A = +2\hat{i} \text{ ms}^{-1}$$

Velocity of particle B,

$$v_B = v_y = -8 \times 3t^2$$

At $t = 1$ s,

$$\Rightarrow v_B = -24\hat{j} \text{ ms}^{-1}$$

Speed of particle B w.r.t. particle A,

$$v_{BA} = -2\hat{i} - 24\hat{j}$$

$\Rightarrow$ Magnitude of velocity,

$$|v_{BA}| = \sqrt{(2)^2 + (24)^2}$$

$$\text{But given } |v_{BA}| = \sqrt{v}$$

$$\Rightarrow v = 580 \text{ ms}^{-1}$$

31. (3)

Sol. The horizontal distance covered by the ball,

$$R = u \cdot T = 20T$$

$\therefore$ The speed of the man to reach the ball,

$$v = \frac{R}{T} = \frac{20 \cdot T}{T} = 20 \text{ m/s}$$

32. (4)

Sol. Let after time t shell hits the plane,

$$\text{then } u_x t = vt$$

[Here, u_x is horizontal component]

$$u \sin \theta t = vt$$

$$\sin \theta = \frac{v}{u} = \frac{200}{400}$$

$$\theta = 30^\circ$$

33. (3)

Sol. Maximum height,

$$H = \frac{u^2 \sin^2 \alpha}{2g}$$

For same speed of projection,

$$H \propto \sin^2 \alpha$$

$$\therefore \frac{H_1}{H_2} = \frac{\sin^2 \alpha}{\sin^2 (90^\circ - \alpha)}$$

$$= \frac{\sin^2 \alpha}{\cos^2 \alpha} = \tan^2 \alpha$$

34. (1)

Sol. The particle will move in a parabolic path till acceleration (due to gravity) is constant. For this the particles should be near the surface of the Earth and air resistance should be negligible

35. (1)

Sol. At highest point of projectile, possesses velocity only along the horizontal direction and is perpendicular to acceleration.

36. (3)

Sol. Statement I is correct but II is incorrect and it can be corrected as.

The acceleration of an object in uniform circular motion is always directed towards the centre of the circle

37. (1)

Sol. Time of ascent

$$= \frac{u \sin \theta}{g} = \frac{15 \sin 37^\circ}{9.8} = 0.92 \text{ s}$$

$$H = \frac{u^2 \sin^2 \theta}{2g} = \frac{(15)^2 \sin^2 37^\circ}{19.6} = 4.16 \text{ m}$$

$$R = \frac{u^2 \sin 2\theta}{g} = \frac{(15)^2 \sin 74^\circ}{9.8} = 22.02 \text{ m}$$

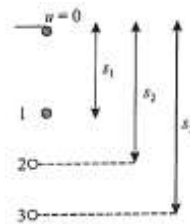
$$T = \frac{2u \sin \theta}{g}$$

$$= \frac{2 \times 15 \sin 37^\circ}{9.8} = 1.84 \text{ s}$$

Hence, (A)→(3), (B)→(1), (C)→(4), (D)→(2)

38. (3)

Sol. Let the drops fall at an interval of t seconds



$$s_1 = \frac{1}{2} g t^2$$

$$s_2 = \frac{1}{2} g (2t)^2 = 4(s_1)$$

$$s_3 = \frac{1}{2} g (3t)^2 = 9(s_1)$$

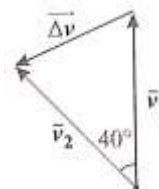
Thus, separation between 3 successive drops below the roof are in ratio

$$s_1 : s_2 - s_1 : s_3 - s_2$$

$$s_1 : 3s_1 : 5s_1 = 1 : 3 : 5$$

39. (4)

Sol. Change in velocity is given as



from figure shown as $|\vec{r}_{\perp 1}| = |\vec{r}_{\perp 2}| = v$ we were

$$|\Delta \vec{v}| = 2v \sin 20^\circ.$$

40. (2)

Sol. velocity of bird

$$v = |t - 2| \text{ m/s}$$

$$\frac{dx}{dt} = t - 2$$

$$dx = (t - 2)dt$$

Integrating both sides,

$$\int_0^x dx = \int_0^4 (t - 2)dt$$

$$x = \left[\frac{t^2}{2} - 2t \right]_0^4$$

$$x = 8 - 8 = 0$$

The displacement of bird is zero

Let time after which bird changes its direction of motion is t_1

$$v = 0$$

$$\text{i.e., } t = 2 \text{ s}$$

$$\text{Distance, } D = \left| \int_0^2 (t-2) dt \right| + \left| \int_2^4 (t-2) dt \right|$$

$$D = 2 + 2$$

$$D = 4 \text{ m}$$

41. (1)

Sol. Error in length,

$$\Delta L = 1 \text{ mm}$$

$$= 1 \times 10^{-3}$$

$$\therefore T = 2\pi \sqrt{\frac{L}{g}}$$

$$\Rightarrow \frac{\Delta g}{g} = \frac{\Delta L}{L} + \frac{2\Delta T}{T}$$

$$= \frac{10^{-3}}{1} + \frac{2 \times 10^{-2}}{1.95}$$

$$= 11.25 \times 10^{-3} \approx 1.13 \times 10^{-2}$$

$$\therefore \Delta g/g \times 100 \approx 1.13 \%$$

42. (3)

Sol. Reading of main scale = $2 \times 1 \text{ mm} = 2 \text{ mm}$

$$\text{Least count of circular scale} = \frac{1 \text{ mm}}{50}$$

$$\text{So, reading of circular scale} = 30 \times \frac{1}{50} \text{ mm}$$

$$= 0.6 \text{ mm}$$

$$\text{Hence, reading} = 2.6$$

So, zero error = 2.6 mm

43. (1)

Sol. Electric flux

$$\phi_E = \vec{E} \cdot \vec{S} \text{ or } \phi_E = E \cdot S = \frac{V}{d} \cdot S$$

$$\therefore \text{Unit of } \phi_E = \frac{\text{volt} \times \text{meter}^2}{\text{metre}} = \text{volt-metre}$$

44. (2)

$$\text{Sol. } \vec{A} + \vec{B} = 3\hat{i} + 3\hat{j} \quad \dots(i)$$

$$\vec{A} - \vec{B} = 3\hat{i} - \hat{j} \quad \dots(ii)$$

Equation (i) + Equation (ii)

$$2\vec{A} = 6\hat{i} + 2\hat{j} \Rightarrow \vec{A} = 3\hat{i} + \hat{j}$$

$$\text{and } \vec{B} = \hat{i} + 2\hat{j}$$

Now, projection of $\vec{A}$ on $\vec{B} = \vec{A} \cdot \frac{\vec{B}}{|\vec{B}|}$

$$= (4\hat{i} + \hat{j}) \cdot \frac{(\hat{i} + 2\hat{j})}{\sqrt{5}} = \frac{6}{\sqrt{5}} \text{ unit}$$

45. (3)

Sol. Here, the angle between A and B is α .

$\therefore (A + B)$ makes an angle θ_1 with A

$$\therefore \tan \theta_1 = \frac{B \sin \alpha}{A + B \cos \alpha}$$

$$\text{Similarly, } \tan \theta_2 = \frac{B \sin \alpha}{A - B \cos \alpha}$$

$$\text{Now, } \tan \theta_1 - \tan \theta_2 = \frac{-2B^2 \sin \alpha \cos \alpha}{A^2 - B^2 \cos^2 \alpha}$$

$$= \frac{-B^2 \sin 2\alpha}{A^2 - B^2 \cos^2 \alpha}$$

CHEMISTRY

Q.46 (3)

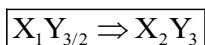
Sol: Mass of $1 e^- = 9.1 \times 10^{-28}$ g.
No. of mole of electron is $10g =$

$$\frac{10}{9.1 \times 10^{-28} \times 6 \times 10^{23}} = \frac{10^6}{9.1 \times 6}$$

Q.47 (4)

Sol:

	X	Y
Wt	75.8	24.2
mole	$\frac{75.8}{75}$	$\frac{24.2}{16}$
	1	$\frac{3}{2}$



Q.48 (1)

Sol: Kilo moles $= \left(\frac{1000}{249.5} \right) = 4$ kilomole

1 mole of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ has 5 mole of H_2O
4 kilo mole $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ has $= (5 \times 4)$ kilomole H_2O
Hence mass $= 5 \times 4 \times 18 = 360\text{kg}$

Q.49 (4)

Sol:-

Q.50 (3)

Sol: Balance reaction



Here Limiting reagent is BO_3
8 mole BO_3 gives 3 mole A_3O_4
1 mole B gives $\frac{3}{8}$ mole A_3O_4

Q.51 (2)

Sol:

Q.52 (4)

Sol: $\frac{M}{M_w} = \frac{5.6}{22.4}$
 $\frac{M}{M_w} = \frac{1}{4}$
 $M_w = 4M$

Q.53 (3)

Sol: $2\text{H}_2 + \text{O}_2 \longrightarrow 2\text{H}_2\text{O}$

2 Vol. 1 vol. 2 vol.

H_2 is limiting reagent

30 ml of H_2 would react with $\text{O}_2 = \frac{1}{2} \times 30 = 15$ ml

So O_2 left $= 20 - 15 = 5$ ml

Q.54 (1)

Sol. Isoelectronic means species having same no. of e^-

Q.55 (3)

Sol: $4f \Rightarrow n = 4, m = -3 \text{ to } +3$
 $\ell = 3 \quad s = \pm 1/2$

Q.56 (4)

Sol.

Q.57 (2)

Sol: Balmer $n_1 = 2$

$n_2 = n$

$$\bar{\nu} = R \left[\frac{1}{2^2} - \frac{1}{n^2} \right]$$

$$R \left[\frac{[n^2 - 4]}{4n^2} \right] = \frac{R(n-2)(n+2)}{4n^2} \text{ cm}^{-1}$$

Q.58 (3)

Sol: $n = 3$

$\ell = 0 \text{ to } n - 1$

$= 0 \text{ to } 3 - 1$

$= 0 \text{ to } 2$

$\boxed{\ell = 0, 1, 2} \rightarrow 3$ subshell

$\boxed{s, p, d}$

No. of orbitals $= n^2$

$= (3)^2$

Q.59 (4)

Sol: $n = 4$

$$\ell = 0, 1, 2, 3$$

$$\ell = 0 \quad m_0 = 0 \quad \rightarrow \text{No } e^-$$

$$\ell = 1 \quad m_1 = -1, 0, +1 \quad \rightarrow 2e^-$$

$$\ell = 2 \quad m_2 = -2, -1, 0, +1, +2 \rightarrow 2e^-$$

$$\ell = 3 \quad m_3 = -3, -2, -1, 0, +1, +2, +3 \rightarrow$$

$$2e^-$$

$$\text{For } m = +1 \rightarrow 6e^-$$

60. (1)

Sol. (a) 1 mol water has molar mass 18 g/mol, since the density of water is approximately 1 g/ml. Therefore, 18g of water occupies a volume of 18 ml.

(b) 1 g molecule of $O_2 = 1$ mol of O_2 which occupies 22.4 L volume.

$$(c) 12.022 \times 10^{23} \text{ molecule of } N_2H_4 = \frac{12.022 \times 10^{23}}{6.022 \times 10^{23}} = 2 \text{ mol}$$

In one mole $N_2H_4 = 4$ atoms of H

In 2 mole $N_2H_4 = 8$ g atoms of H.

$$(d) 0.16 \text{ g } CH_4 = \frac{0.16}{16} = 0.01 \text{ mol}$$

In 1 mole $CH_4 = 6$ mol neutrons

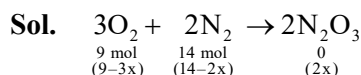
In 0.01 mole $CH_4 = 0.06$ mol neutrons

61. (3)

Sol. $dH_2O = 1 \text{ g/mL}$

$$\text{So, } W_{H_2O} = 1000 \text{ g} \Rightarrow 55.55 \text{ moles}$$

62. (3)



According to question

$$n_{O_2} = 3 \text{ mol}$$

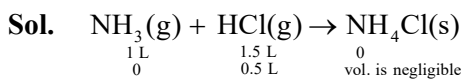
$$(9 - 2x) = 3$$

$$x = 2 \text{ mol}$$

$$\text{So, moles of } N_2O_3 = 2x$$

$$= 4 \text{ mol}$$

63. (2)



$\therefore$ After reaction volume is 0.5 L.

64. (1)

Sol. In Fe_2O_3 :

112 g Fe reacts with 48 g O.

$\therefore$ 56 g Fe reacts with 24 g O.

In FeO :

56 g Fe reacts with 16 g O.

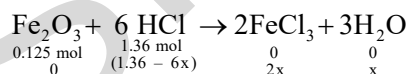
$\therefore$ Ratio of masses of O = 24 : 16 = 3 : 2

65. (2)

$$\text{Sol. Moles of } Fe_2O_3 = \frac{20}{160} = 0.125 \text{ mole.}$$

$$\text{Moles of } HCl = \frac{50}{36.5} = 1.36 \text{ mole}$$

Fe_2O_3 is limiting reagent



Remaining moles of HCl

$$= 1.36 - 6 \times 0.125$$

$$= 0.61 \text{ mol HCl}$$

$$\therefore \text{Mass} = 0.61 \times 36.5 \text{ g}$$

$$= 22.265 \text{ g HCl}$$

66. (3)

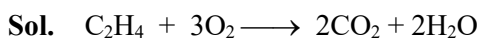
Sol. 1.06

In 1 mole $(NH_4)_3 PO_4$

For 12 mol H-atoms – 4 mol O atoms are present.

$\therefore$ For 3 mol H-atoms – 1 mol O - atoms

67. (1)



C_2H_4 & O_2 are in 1 : 3 vol.ratio

i.e O_2 will be 60 mL.

$$\therefore V_{air} \approx 5 \times 60 \approx 300 \text{ mL}$$

68. (3)

Sol. $PV = nRT$

$$PV = \frac{M}{M_w} RT$$

$$M_w = \frac{MRT}{PV}$$

$$\frac{0.22}{0.112} \times 0.0821 \times 273$$

$$M = 44 \text{ g}$$

$$\text{NO}_2 = 46 \text{ g}$$

$$\text{N}_2\text{O} = 44 \text{ g}$$

$$\text{CO}_2 = 44 \text{ g}$$

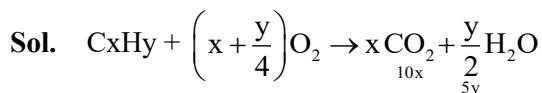
$$\text{C}_3\text{H}_8 = 44 \text{ g}$$

69. (3)

$$\text{Sol. } 0.4 = \frac{1 \times 56 \times 100}{\text{min. } M_w}$$

$$\text{Min. } M_w = 14000$$

70. (1)



$$10x = 40$$

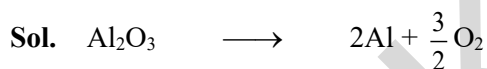
$$x = 4$$

$$5y = 50$$

$$y = 10$$



71. (4)



1 mole

2 mol

$$102 \text{ g} \quad 54 \text{ g}$$

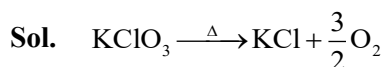
170 g of pure sample of Al_2O_3 will yield

$$\frac{53.98}{101.96} \times 170 = 90 \text{ g Al}$$

$$\text{percentage purity} = \frac{\text{actual mass}}{\text{theoretical mass}} \times 100$$

$$= \frac{18}{90} \times 100 \approx 20\%$$

72. (1)



$$n_{\text{KClO}_3} = \frac{n_{\text{O}_2}}{3/2} \Rightarrow \frac{1/4}{3/2} \Rightarrow \frac{1}{6}$$

73. (2)

Sol.

$$\text{I. } n = 4, \ell \quad \zeta, s = -\frac{1}{2}; \text{ represents}$$

$$4d (n + \ell = 6)$$

$$\text{II. } n = 3, \ell \quad \zeta, s = +\frac{1}{2}; \text{ represents}$$

$$3d (n + \ell = 5)$$

$$\text{III. } n = 4, \ell \quad \zeta, s = +\frac{1}{2}; \text{ represents}$$

$$4p (n + \ell = 5)$$

$$\text{IV. } n = 3, \ell \quad \zeta, s = +\frac{1}{2}; \text{ represents}$$

$$3p (n + \ell = 4)$$

Order of energy depends on the $(n + \ell)$, greater is the $(n + \ell)$ value greater is the energy, if $(n + \ell)$ is same, then it depends on n ; if 'n' is more, energy is more.

According to $(n + \ell)$

$$\text{Energy} = \text{I} > \text{III} > \text{II} > \text{IV}$$

74. (1)

$$\text{Sol. } E_n = -R_H \left(\frac{Z^2}{n^2} \right)$$

For $\text{He}^+ (n = 1)$

$$E_n = -x$$

$$= -R_H \left(\frac{2^2}{1^2} \right)$$

$$= -4R_H$$

$$R_H = \frac{x}{4}$$

For $\text{Be}^{3+} (n = 2)$

$$E_n = -R_H \left(\frac{Z^2}{n^2} \right) \text{ J}$$

$$= \frac{-x}{4} \times \left(\frac{4 \times 4}{2 \times 2} \right)$$

$$= -x \text{ J}$$

75. (3)

Sol. $\Delta x \cdot \Delta p = \frac{h}{4\pi}$ or $\Delta x \cdot m\Delta v = \frac{h}{4\pi}$

$$\Delta v = \frac{0.011}{100} \times 3 \times 10^4 = 3.3 \text{ cms}^{-1}$$

$$\Delta x = \frac{6.6 \times 10^{-27}}{4 \times 3.14 \times 9.1 \times 10^{-28} \times 3.3}$$

$$= 0.175 \text{ cm}$$

76. (2)

Sol. $V \propto \frac{Z}{n}$

77. (1)

Sol. $\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

78. (2)

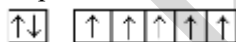
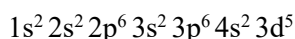
Sol. Fact

79. (1)

Sol. $\Psi^2 = 0$ means probability of finding the e^- in the orbital is zero. It represents a node.

80. (2)

Sol. Mn $\rightarrow$ Atomic No. = 25



81. (3)

Sol. We know that, $E = \frac{hc}{\lambda}$

Let, power = P

then, energy in time t

$$nE = Pt \text{ and } n = \frac{Pt}{E} = \frac{Pt\lambda}{hc}$$

$$\therefore n_R = \frac{Pt\lambda_R}{hc} \quad \dots(i)$$

$$n_g = \frac{Pt\lambda_g}{hc} \quad \dots(ii)$$

Dividing Eq. (i) by Eq. (ii), we get

$$\frac{n_R}{n_g} = \frac{\lambda_R}{\lambda_g}$$

We know, $\lambda_R > \lambda_g$

$$n_R > n_g$$

82. (1)

Sol. $E = n \frac{hc}{\lambda}$

$$E = n \times \frac{hc}{\lambda}$$

$$10^{-7} = n \times \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{5000 \times 10^{-10}}$$

$$n = 2.5 \times 10^{11} \text{ photons per second}$$

83. (3)

Sol. $R_H \propto \text{mass of electron}$

$$R_H = \frac{m_e e^4}{8 \epsilon_0^2 h^3 C}$$

$$R_H \propto m_e$$

$$m'_e = m_e - 0.25m_e$$

$$= 0.75m_e$$

84. (3)

Sol. Only few around one in 20,000 particles were bounce back.

85. (3)

Sol. Orbital having angular node (ℓ) = 3

Total node = Radial node + angular node

$$= n - \ell - 1 + \ell$$

$$3 = n - 1$$

$$n = 4$$

$$\text{Subshell "n\ell" = 4f}$$

86. (3)

Sol. $\text{Fe}^{+2} \Rightarrow (\text{Ar})4s3d^6$; d-electron $\Rightarrow 6$

$\text{Cl } 1s^2 2s^2 2p^6 3s^2 3p^5$; p-electron = 11

87. (3)

Sol. $h\nu = h\nu_0 + \text{KE}$

or

$$E = E_0 + \text{KE}$$

$$6.2 = 4.2 + \text{K.E}$$

$$\text{K.E} = 2\text{eV}$$

or

$$2 \times 1.6 \times 10^{-19} \text{ J}$$

$$= 3.2 \times 10^{-19} \text{ J}$$

88. (2)

Sol. Cathode rays consist of electrons which are the basic constituent of all the atom and hence their characteristics do not depend upon the material of electrodes and nature of the gas present in the cathode ray tube.

89. (3)

Sol. According to de Broglie equation, $\lambda = \frac{h}{mv}$,

hence wave length is inversely proportional to mass of a particle and hence can be observed only for microscopic particles with very small mass.

90. (4)

Sol. One mole of any substance corresponding to 6.023×10^{23} entities is respective of its weight.
Molecular weight of $\text{SO}_2 = 32 + 2 \times 16 = 64\text{gm}$.
Molecular weight of $\text{O}_2 = 16 \times 2 = 32\text{gm}$.
 $\therefore$ Molecular weight of SO_2 is double to that of O_2 .